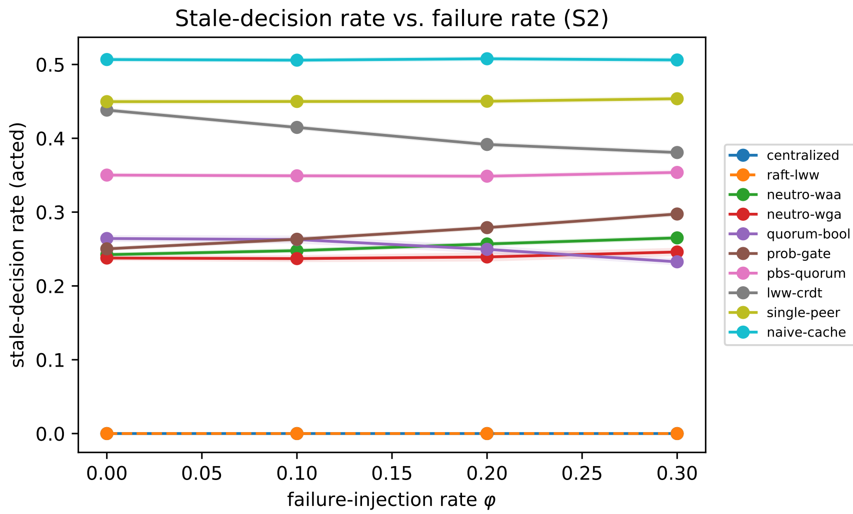

Figure S1: Stale-decision rate vs. failure-injection rate (S2). Bands are 5000-resample bootstrap 95% CIs.

Supplementary Information

A decentralized neutrosophic decision layer for global-state freshness in microservices architectures

Haitham A. El-Ghareeb

Supplementary Figures

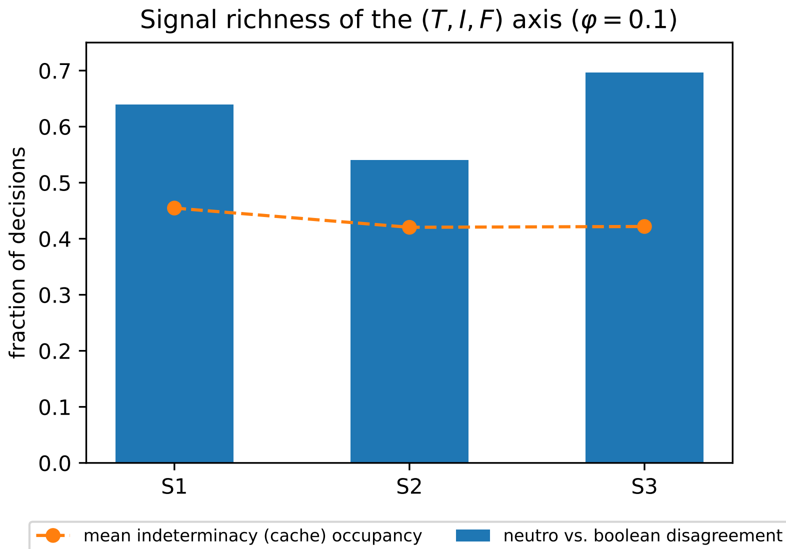

Figure S2: RQ1: the indeterminacy axis is well populated and the (T, I, F) decision diverges from a boolean quorum on a substantial fraction of decisions.

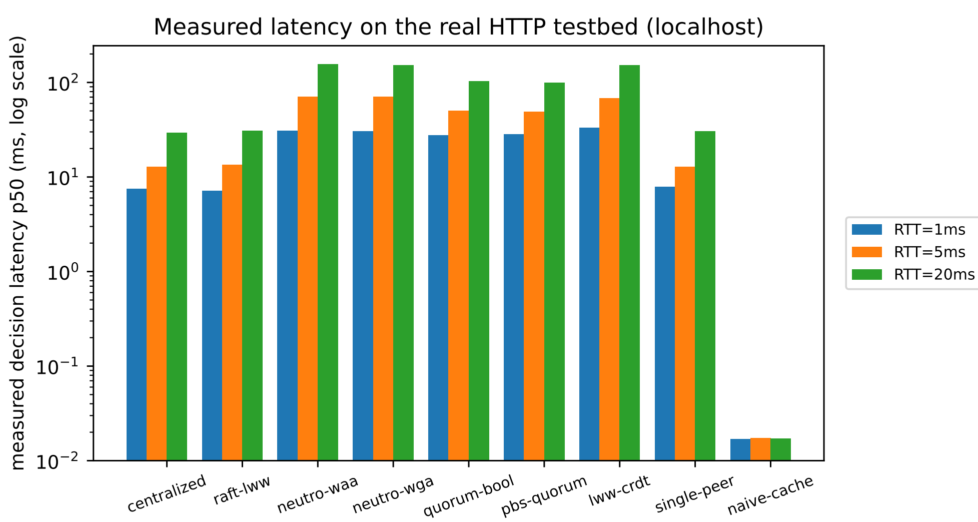

Figure S3: Measured median decision latency on the real localhost HTTP testbed: local read, single round-trip, and R-peer fan-out tiers. Log scale: naive-cache is a local read (≈ 0.02 ms), dwarfed by the network tiers but fully measured ($n = 200$ per cell).

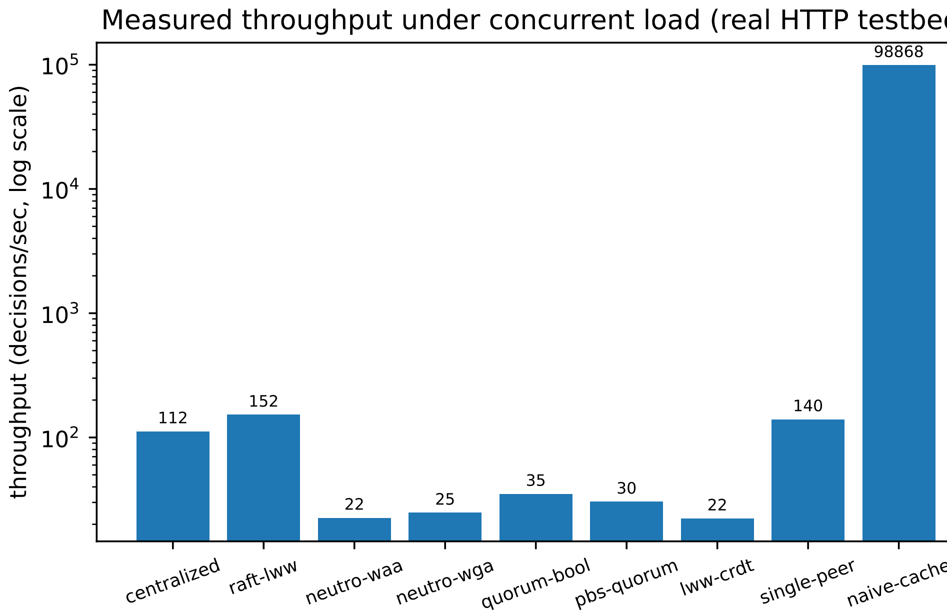

Figure S4: Measured throughput (decisions/sec) under concurrent closed-loop load on the real HTTP testbed.

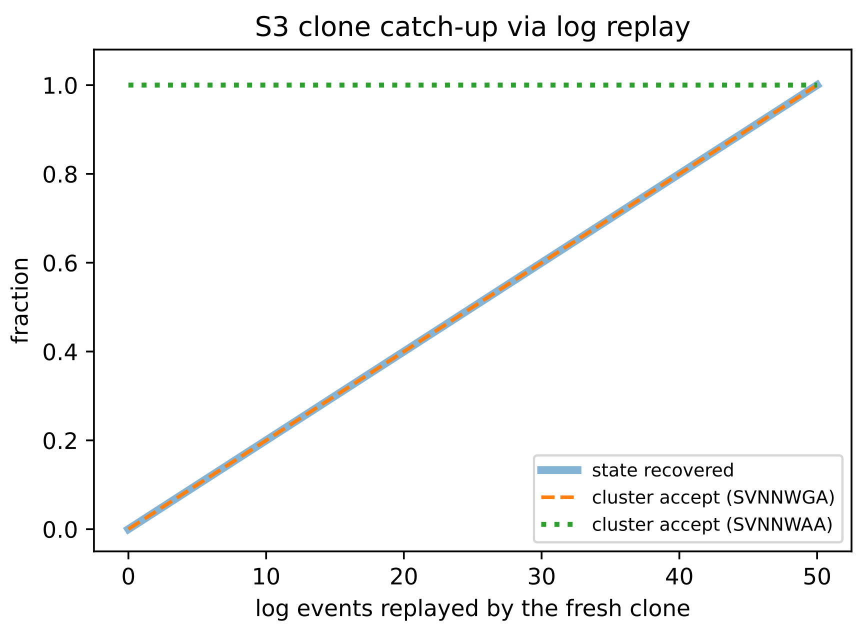

Figure S5: S3: a fresh clone recovers state by replaying the log; SVNNWGA acceptance trails until catch-up while SVNNWAA stays available throughout. State-recovery and SVNNWGA acceptance coincide (dashed line over the thick one).

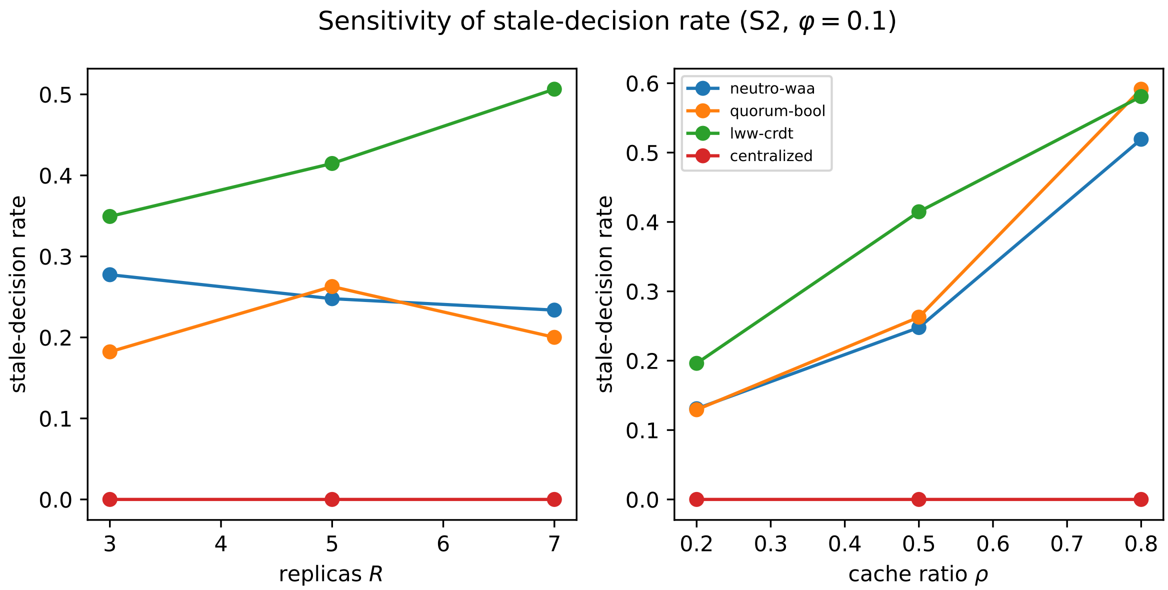

Figure S6: Sensitivity of the stale-decision rate to replica count R and cache ratio ρ (S2, $\varphi = 0.1$).

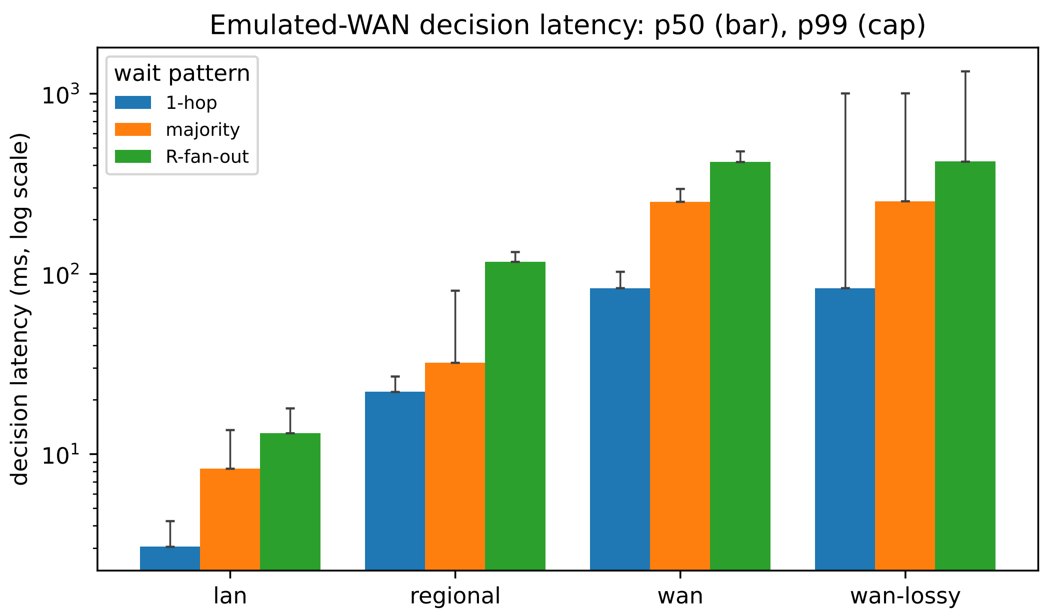

Figure S7: Decision latency on a single-host Docker deployment under *emulated* WAN (Toxiproxy per-link latency, jitter, and loss): p50 (bar) and p99 (cap), log scale. The wait patterns (single-hop, majority, R-peer fan-out) separate cleanly and the gap grows with link latency; under 2% loss the tails inflate to the 1 s decision deadline, most for the fan-out (it waits for the slowest of R peers). Real HTTP over proxied sockets; not geo-distributed.

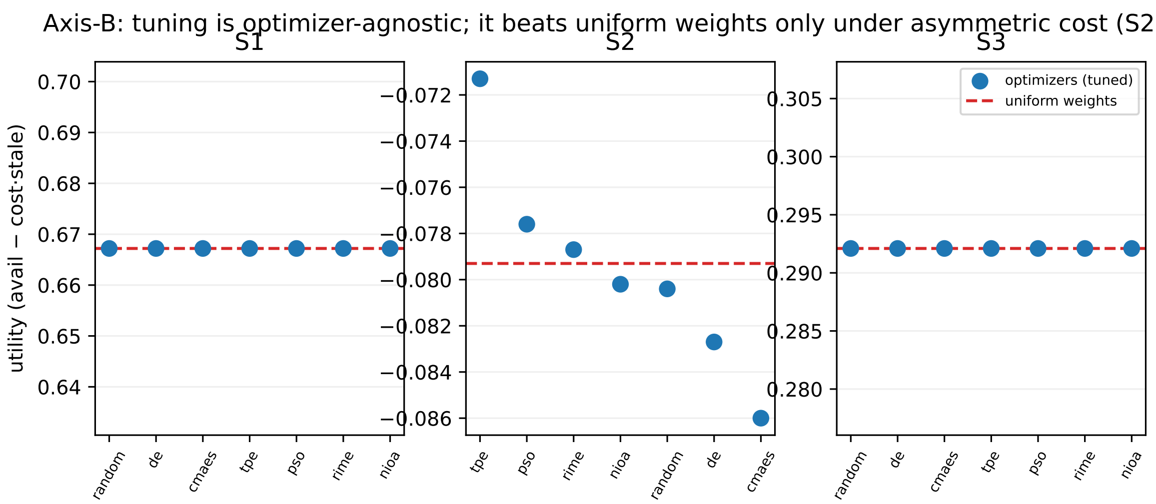

Figure S8: Axis B: equal-budget metaheuristic tuning of $[\tau, \text{weights}]$ (seven optimizers, 120 evaluations each), tuned utility per optimizer (points) versus the uniform-weights baseline (dashed). At equal budget the optimizers converge—identically on the symmetric S1/S3, within ≈2% on the asymmetric S2—and tuning beats uniform weights only on S2 (+10.1%); on S1/S3 uniform is already optimal. No-Free-Lunch: the gain is not an artefact of the optimizer.

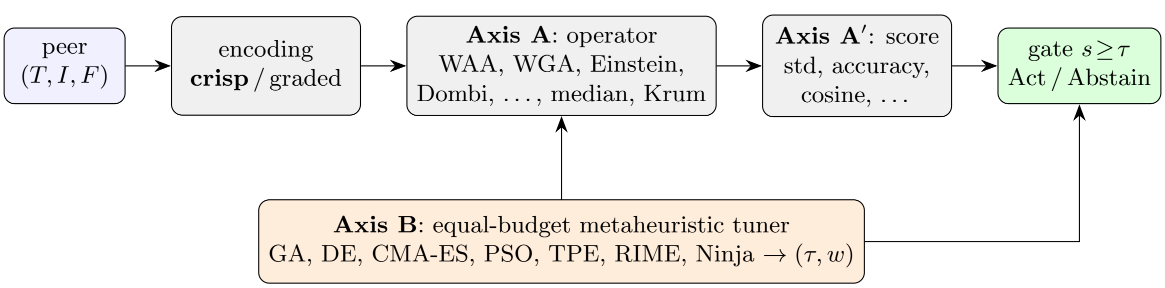

Figure S9: The two-axis audit pipeline. Peer views are encoded (crisp anchor or graded), aggregated by an operator-panel member (Axis A), deneutrosophized by a score-panel member (Axis A′), and gated at the calibrated threshold; an equal-budget metaheuristic panel (Axis B) tunes the threshold and peer weights. One axis is varied at a time, holding the others at the submitted anchor.

Table S1: Graded-encoding anchors (the GRADE map of Algorithm 2). A peer's observable kind and version-lag ℓ relative to the most up-to-date reachable peer (freshness $\varphi = 1/(1+\ell/\sigma)$, softness $\sigma = 2$) map to a single-valued neutrosophic triple (T, I, F): T is confidence the held value is the committed-current truth, I is unconfirmedness, F is confidence the value is stale, superseded, or absent. The anchors are a documented, tunable design choice (calibration override `grade_encoding`); the crisp corners $(1,0,0)/(0,1,0)/(0,0,1)$ remain the conservative baseline.

Peer kind (observable)	T	I	F
Persisted, fresh ($\ell = 0$)	1.00	0.10	0.00
Persisted, lagging ($\ell \to \infty$)	0.50	0.10	0.40
Clean cache, fresh	0.45	0.55	0.10
Clean cache, lagging	0.00	0.65	0.35
Suspected dirty write	$0.15\,\varphi$	0.45	0.45
Absent / unreachable	0.05	0.10	0.85

Supplementary Tables

Table S2: Headline results for S2 at $\varphi = 0.1$ (30 reps/cell, 3 seeds; stale-rate with 5000-resample bootstrap 95% CI). All cells `validated`.

System	Partition	Stale-rate [95% CI]	Availability	Msgs/dec
Centralized	none	0.000 [0.000, 0.000]	0.900	1
Raft-LWW	none	0.000 [0.000, 0.000]	0.900	1
Neutro-WAA	none	0.248 [0.244, 0.251]	0.942	5
Neutro-WGA	none	0.237 [0.232, 0.242]	0.427	5
Quorum-bool	none	0.263 [0.257, 0.268]	0.402	5
PBS-quorum	none	0.349 [0.346, 0.353]	1.000	3
LWW-CRDT	none	0.415 [0.411, 0.418]	1.000	5
Single-peer	none	0.450 [0.446, 0.453]	1.000	1
Naive-cache	none	0.506 [0.503, 0.509]	1.000	0
Centralized	transient	0.000 [0.000, 0.000]	0.453	1
Raft-LWW	transient	0.000 [0.000, 0.000]	0.453	1
Neutro-WAA	transient	0.263 [0.260, 0.266]	0.880	5
Neutro-WGA	transient	0.240 [0.235, 0.245]	0.332	5
Quorum-bool	transient	0.223 [0.218, 0.227]	0.410	5
PBS-quorum	transient	0.348 [0.345, 0.351]	0.996	3
LWW-CRDT	transient	0.380 [0.377, 0.384]	0.996	5
Single-peer	transient	0.450 [0.446, 0.453]	0.996	1
Naive-cache	transient	0.506 [0.503, 0.510]	1.000	0

Table S3: Headline results for S1 at $\varphi = 0.1$ (30 reps/cell, 3 seeds; stale-rate with 5000-resample bootstrap 95% CI). All cells `validated`.

System	Partition	Stale-rate [95% CI]	Availability	Msgs/dec
Centralized	none	0.000 [0.000, 0.000]	0.901	1
Raft-LWW	none	0.000 [0.000, 0.000]	0.901	1
Neutro-WAA	none	0.137 [0.134, 0.139]	1.000	5
Neutro-WGA	none	0.106 [0.103, 0.109]	0.427	5
Quorum-bool	none	0.106 [0.102, 0.109]	0.361	5
PBS-quorum	none	0.272 [0.269, 0.275]	1.000	3
LWW-CRDT	none	0.343 [0.340, 0.347]	1.000	5
Single-peer	none	0.327 [0.324, 0.331]	1.000	1
Naive-cache	none	0.386 [0.382, 0.390]	1.000	0
Centralized	transient	0.000 [0.000, 0.000]	0.452	1
Raft-LWW	transient	0.000 [0.000, 0.000]	0.452	1
Neutro-WAA	transient	0.171 [0.168, 0.174]	0.997	5
Neutro-WGA	transient	0.109 [0.105, 0.114]	0.318	5
Quorum-bool	transient	0.089 [0.085, 0.092]	0.370	5
PBS-quorum	transient	0.269 [0.266, 0.272]	0.997	3
LWW-CRDT	transient	0.306 [0.303, 0.309]	0.997	5
Single-peer	transient	0.327 [0.324, 0.331]	0.997	1
Naive-cache	transient	0.388 [0.384, 0.391]	1.000	0